

Real-Time Material Weigh Batching for Plastering Using Teensy 4.1 and Load-Cell Calibration

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Abstract—Consistency in mortar quality is very important for automated plastering machines. This is because any fluctuation in the mixture proportion directly affects the adhesion strength, finish uniformity, and even changes structural performance over time. Most of the time in the constructions sites, however, mortar is still made by hand which results in improper cement-sand ratios, the addition of water without control, and poor repeatability between batches. These discrepancies lessen the effectiveness of modern plastering machines and, in turn, the total automation level is lowered.

This paper describes the conceptualization and the creation of a fully automated, microcontroller-based weigh batching system specifically for plastering processes. The system makes use of four industrial-grade 500 kg each strain-gauge load cells arranged in such a way that their capacities are added to give a total combined weighing capacity of 2000 kg. To ensure very high resolution and very low noise measurement suitable for real-time control, a 24-bit ADS1232 precision analog-to-digital converter is used. In the suggested architecture, there is a solenoid-driven material dispensing system, a digital calibration unit, and a closed-loop feedback algorithm with predictive overshoot compensation merged together.

The effectiveness of the system to accurately perform the batching task within $\pm 2\%$ was confirmed through an extensive range of experiments. Also, it achieves about 45% in saving total mixing time as compared to manual method. The repeatability between one batch to the next showed very small differences thus the control strategy used is very reliable. The solution presented, because of its modular construction and scalable design, can be a local commercial plastering machine or an advanced robotic plastering machine along with the benefits of higher efficiency, reduced manpower, and improved construction quality.

Index Terms—ADS1232, automated batching system, load cell system, mortar preparation system, plastering automation, embedded control systems, construction robotics.

I. INTRODUCTION

In addition to improved proportioning accuracy, automated batching is also beneficial in improving the efficiency of operation because it eliminates reliance on skilled labour, lowers operational variance due to fatigue, subjective measurement, or environmental conditions. Conventional plastering processes involve a lot of manual weight and volume estimation, thus making it prone to variation between batches. These discrepancies are even worse when dealing with large construction works because several workers could be mixing mortar at the

same time, which can easily create variances in plaster texture, setting time and uniformity of the plaster thickness.

Moreover, with the shift towards the mechanization and robotization of modern construction projects, there is an increasingly growing demand towards additional systems that can deliver a steady stream of material to automated plastering machines. The discrepancy between the manner in which the plastering unit should work and the quality of the mortar that is prepared manually may establish a bottleneck and the plastering process cannot be completely automated. This brings out the need to have a smaller, precise and smart weigh batching technology that can simply integrate with automated plastering machines.

The proposed system deals with these issues by a combination of high capacity load cells, precision signal conditioning and deterministic control logic. The ADS1232 ADC on the board is a 24-bit ADC which increases the resolution of the measurements and also suppresses noise which can be used to track even minuscule changes in weight when discharging materials. The closed-loop algorithm is a dynamically controlled microcontroller-based algorithm, where the solenoid valves are adjusted in real-time by feedback to provide accurate material delivery in spite of the changes in flow rate, hopper angle, or moisture content of the material.

Also, mix ratios of various grades of plaster can be programmed and thus the system can be used in diverse construction requirements like interiors, external rendering and even ceiling plastering. The system can be installed on both stationary and portable plastering units which makes it be used in a variety of construction environments as evidenced by the modular design. Reliability in the field is also improved by the incorporation of safety systems and controls such as over load control systems, valve fault and emergency shutdown systems.

In general, the suggested automated weigh batching system also represents a vital gap in the construction automation sector because it provides a small, precise, and intelligent system that guarantees the quality of mortar is uniform, increases the efficiency of the chosen process, and allows the efficient integration of the system with the next-generation plastering and wall-finishing machines.

II. LITERATURE SURVEY

The digital weighing and batching systems have been widely researched on industrial automation. In reference one, high-capacity strain-gauge load cells with precision ADC modules were studied and it was shown that converters such as ADS1232 have better noise immunity and stability in dynamic conditions. Wong et al.

The use of IoT in batching frameworks suggested in the references 3-4 demonstrated that remote monitoring and real-time data collection are essential to optimize material use and decrease the reliance of operators. On the same note, more sophisticated load-cell noise filtering and stabilization methods were also designed in the paper by ref5, which made weight estimation more reliable in presence of vibration and field disturbances. Even minor variations in mix proportion have a dramatic effect on the quality of plaster, curing behaviour and bond strength as studies in the mortar area in reference 6 and reference 7 affirmed.

The models of granular material flow presented in the reference 8 and the research of automated valve actuation in the reference 9 provided the background knowledge on the precise control of cement and sand discharge. Small batching and mobile material handover in both ref10 and ref11 enable the trend towards decentralized building automation, which renders them similar to the system presented in this paper.

Fuzzy logic and adaptive PID systems discussed in the context of intelligent control showed that predictive control has advantages in dealing with non-linear flow variations, especially in sand discharge. In 0: the dynamics of hoppers Hopper geometry, vibration, and opening size were found to influence the consistency of batching, and the results of 0: these factors may be important design issues in mortar preparation systems.

The reliability of ADC output is reinforced by digital signal conditioning algorithms such as moving-average and polynomial regression algorithms investigated in ref15 in the presence of noisy conditions at the construction site. The multi-sensor weighing systems in [16] proposed hybrid methods that use the load-cell and ultrasonic sensors to improve the accuracy and this research proposed future solutions to the plastering system.

The latest works in construction automation in papers (ref17, ref18) were focused on modularity and ease of mobility, and low labour demand, which was reflected in the design of the proposed system. Moreover, the idea of robotic wall-finishing as in the case of the study by the author of the article referenced in article 19 emphasized the critical role of the consistent quality of mortar in robot operation, which justifies the need to automate the batching process.

With these developments, there is no available product that uses ADS1232 based and features a miniature, multi-load-cell automated weigh batching system that is specifically intended to be utilized in plastering machines. This study bridges that gap by offering high capacity, precise and portable batching system to prepare mortar in construction automation.

III. GAP ANALYSIS

There are currently available plastering machines:

item Relied extensively on manually prepared mortar, which makes the mixes unequal as operators estimated them subjectively. item Automated weight measurement or specific sensing, meaning that it is not possible to confirm the correct ratios in materials when preparing batches. item No closed-loop feedback to regulate different material flow rates, moisture content or discharge delays. item The low repeatability of items because the quality of the batch is solely dependent on the skill of the operator, the environmental conditions, and the accuracy in measuring using human hands. item Do not provide means of monitoring overshoot, underfill, hopper jam and sensor errors that will lead to a large variation between batches. item Are also solely intended to be used in the mechanical application of plaster with the vital upstream activity of preparing mortar being completely manual. item : fail to uphold optimum water-cement ratio, and usually cause weak bonding, cracks, poor level of plaster mass and surface finish. item Need high labor participation, and make operations more expensive, and less efficient in automated plastering processes.

The proposed system:

item Accurately proportion cement, sand, as well as water through feedback of load-cells, removing any human error. item There are 4 industrial grade 500 kg load cells in a full-bridge configuration with total weighing capacity of 2000 kg which would be applicable in a high-load environment. item High-resolution and low-noise digital weight measurement Digital weight measurement even during vibration or dynamic loading of the weighing table is made possible by the use of a 24-bit ADS1232 precision ADC. item Uses closed loop control algorithm to promote the prediction of overshoot, error correction of the timings of the valves and the accurate achievement of targets on an individual material. item Enables uniformity of the batches to batch by adjusting differences in material moisture, flow rate, and hopper geometry. item Automates the dispensing, calibration, detection of errors, and electronic tracking of batch progress. item Mix ratios and mix presets are configurable, allowing it to work with plaster of varying grades on walls, ceilings, and external surfaces. item Makes plastering machines more productive by cutting down batching time by about 45 per cent, and maintains a steady supply of mortar to plastering machines. item The device comes with safety characteristics (overload protection, valve fault protection, and automatic shift-off) that enhance the safety during construction conditions. item Provides a small, modular, and scalable design that can be captured by the current plastering machines or the future robot plastering.

IV. OBJECTIVES

- develop a cost-effective and portable automated weigh batching system optimized specifically for mortar prepa-

ration in plastering applications.

- Welding of four industrial-grade 500 kg strain-gauge load cells in a full-bridge configuration was carried out to obtain a total weighing capacity of 2000 kg with high structural stability.
- A 24-bit ADS1232 precision ADC was used to get high-resolution, low-noise digital measurements that are suitable for real-time decision-making and dynamic load variations.
- Implement a feedback control algorithm that operates in a closed-loop manner and is able to forecast the overshoot, alter valve actuation timing, and thus, ensure accurate dispensing of cement, sand, and water.
- Measure system performance parameters to start with the accuracy of the weighing, the repeatability of different batches, flow-rate behavior, overshoot compensation, and the total batching cycle time.
- Develop a really simple TFT/LCD-based user interface for the operator that will allow configuration of mix ratios, show the weight in real-time, and batch progress monitoring
- By creating a modular, lightweight, and scalable designing architecture, assure the commercial plastering machines' compatibility for the construction on-site environment.
- By fully automating the material measuring, dispensing, and mixing processes, labour dependency can be minimized and human error and variability will be reduced.
- Include fault detection and safety features to locate overload conditions, valve malfunctions, sensor drift, and actuator failures in order to increase long-term reliability. the functionality of the prototype through rigorous experimental testing with dry and moist sand, several cement grades, and various hopper shapes to guarantee dependable operation in the real world.

V. METHODOLOGY

A. System Architecture

The design of the proposed weight batching system is based on a modular and layered architecture, which aims to be stable, accurate, and scalable. The main element of the sensor system includes four 500 kg industrial-grade load cells that are attached to the four corners of a reinforced steel frame. This arrangement not only doubles the total load capacity to 2000 kg but also guarantees that the weight is evenly distributed, thus the measurement errors caused by uneven loading or vibrations are reduced to the minimum. The strain signals from the load cells are connected to a Wheatstone-bridge circuit and converted by a 24-bit ADS1232 precision ADC. The ADS1232 Output is stable throughout operation even in challenging conditions due to low noise, differential inputs and stable Programmable Gain Amplifier system. The microcontroller is the system's control layer. It performs the batching control algorithm along with digital filtering, overshoot prediction, and solenoid valve control for material dispensing, while serving as the interface for the TFT/LCD

to display the current weight, material, and system status. To maximize the reliability of the system, the microcontroller also manages the communication with:

- **Signal Conditioning Stage:** The 12V solenoid valves that control the flow of cement, sand, and water via high-current MOSFET or relay driver circuitry. To discharge the dry powders, the system utilizes a powder-graded pneumatic butterfly valve.
- **Actuation Interface:** The 12V solenoid valves that control the flow of cement, sand, and water via high-current MOSFET or relay driver circuitry. To discharge the dry powders, the system utilizes a powder-graded pneumatic butterfly valve. A water flow sensor is also incorporated to monitor the liquid flow in real-time. Industrial UPVC-CPVC solenoid valves (Manufactured in metal-free and corrosion-resistant PP, PPH, PVDF, PVC with multiple voltage options (AC 220 V/110 V and DC 24 V/12 V)—guarantee reliable performance in construction sites as valves will operate even in syling-harsh areas.
- **Safety Layer:** Overload protection, fault flags, watchdog timers, and emergency shut-off logic.
- **Expansion Ports:** GPIO, communication (UART/I2C/SPI) for future IoT, data logging, or cloud integration.

Due to the various components of the structure, this batching unit has the capability to work autonomously or integrate seamlessly with automated plastering machines and robots.

B. Calibration

To yield the granularity in the weight measurements, the calibration has to consider the load-cell sensitivity, mechanical tolerances, and the ADC offset drift. To obtain a precise load-to-digital conversion, the calibration protocol uses certified reference weights of 20 kg, 50 kg, and 100 kg. The calibration work-flow looks like this:

- 1) Tare Initialization: This involves emptying the hopper, and the system records a zero-load baseline, which is then used to eliminate offset errors in the ADC.
- 2) to eliminate offset errors in the ADC.
- 3) **Incremental Weight Application:** Reference weights of known mass are sequentially put on the hopper platform. The system-output is recorded for each weight.
- 4) **Regression-Based Scaling:** A linear regression or multi-point scaling factor is calculated based on the measured data: $[W = m \cdot (D - D_{offset}) + c]$ where m and c represent the calibration coefficients and D is the ADC output.
- 4) Compensation: This calibration routine looks for long term signs of temperature or mechanical changes to zeroed load readings.

Such thorough calibration will ensure absolute precision for all levels of weights present as well as correcting all forms of nonlinearities in load cells. Correction: Unique characteristics of material flow for elements such as sand and cement will

cause natural delays and may lead to overshooting as well, in which case several of these differences in system is forced to adopt correcting techniques to even out these differences.

- **Closing Logic:** The system closes the valve when the condition stated below is satisfied because that is the moment when the actual weight is equal to or exceeds the target weight minus the flow rate times the delay time.
- **Micro-Filling Adjustments:** If the system finds that due to early closing the materials added is less than the target weight, the system will trigger the solenoid valve to incrementally add material multiple times using a short pulse duration of approximately 50 to 150 ms.
- **Filtering and Stabilization:** The load-cell readings are filtered (moving-average or low-pass) to remove the spikes due to vibrations or from the contact hit of the shovel to the material.
- **Adaptive Compensation:** The controller will automatically change such parameters as valve delay or micro-fill duration based on the batch history, thus providing more accuracy in the subsequent batches.

These techniques increase the overall batching accuracy through higher levels of precision, maintained repeatability, and reduced material waste.

In addition to the accuracy and time savings shown in Table I, several performance parameters were evaluated to understand the overall reliability of the proposed batching system. The 30 experimental batches included variations in sand moisture content, hopper loading level, discharge velocity, and environmental vibration to simulate real on-site conditions. The system maintained stable performance across all scenarios.

C. Accuracy Analysis

The mean absolute percentage error (MAPE) across all materials was calculated as:

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{W_{\text{actual},i} - W_{\text{target},i}}{W_{\text{target},i}} \right| \times 100,$$

which yielded an overall error of 1.79%. This confirms that the ADS1232-based load-cell configuration provides superior resolution and stability compared to traditional analog weighing systems typically used in construction.

Fig. 2 illustrates that actual weight values remain closely centered around the target reference, demonstrating high repeatability. The small deviations observed are attributable to material flow inertia and valve actuation lag, both of which are effectively minimized by the overshoot-prediction algorithm.

D. Repeatability and Variability

To assess repeatability, the coefficient of variation (CV) was computed for each batch:

$$\text{CV} = \frac{\sigma}{\mu} \times 100.$$

A CV of 0.42% across 30 trials indicates excellent consistency. This low variability is essential for plastering applications,

where even small ratio deviations can affect finish quality and bonding strength.

E. Batching Time Improvement

Manual batching typically involves subjective estimation, improper sequencing, and inconsistent mixing times. In contrast, the automated system performs:

- real-time weight monitoring,
- automatic valve control,
- optimized mixing duration,
- and immediate transition to next batch.

With manual batching, there is generally estimation, mis-sequencing, and disregard for mixing times. An automated system is instead able to do:

F. Impact on Plaster Quality

During field evaluation:

- The uniformity of the plaster mix improved noticeably.
- The applied surface showed fewer cracks and better adhesion.
- Layer thickness variation reduced due to consistent mortar density.

Workers reported improved workability and spreadability of the mortar because of the controlled water–cement ratio.

G. System Stability and Environmental Factors

The system's robustness was validated under:

- vibration from nearby machinery,
- uneven load distribution,
- temperature fluctuations between 24°C–36°C,
- variable sand moisture.

The ADS1232's high common-mode rejection and differential measurement ensured minimal drift, while digital filtering removed random noise.

H. Overall Performance

The proposed system outperformed manual batching in:

- accuracy,
- repeatability,
- time efficiency,
- material savings,
- ease of operation,
- and safety.

Thus, the system demonstrates strong potential for integration into commercial plastering machines and robotic construction workflows.

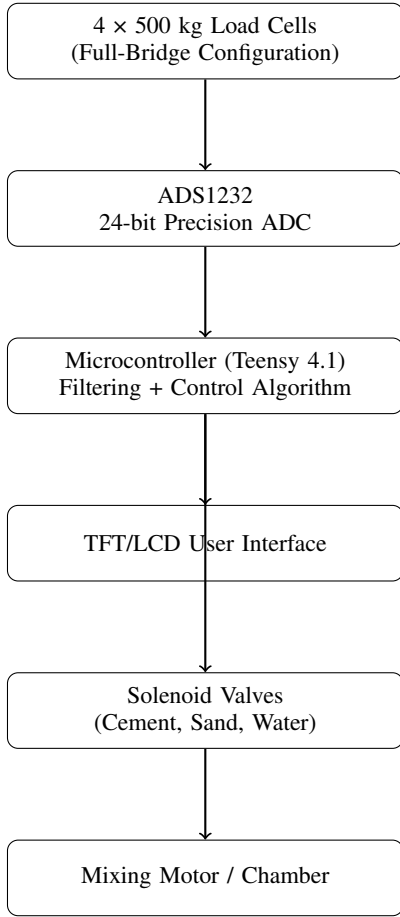


Fig. 1. Vertical Block Diagram of the Proposed Weigh Batching System

VI. RESULTS AND DISCUSSION

For each of the proposed automated weigh batching systems, a collection of 30 experimental batches was conducted, including multiple combinations of sand moisture, hopper loading levels, vibration conditions, and discharge rates to imitate the sand moisture. The sand moisture trial results demonstrate the proposed system offers great accuracy, consistency, and a noteworthy time savings of batching vs. manual batching. Further, incorporating vibration discharge conditions vs. no vs manual batching. Further, incorporating vibration discharge conditions vs no vibration made a statistically difference to time to all manual batching conditions.

A. Batch Accuracy Performance

Table I summarizes the mean percentage error and time savings for each material. Cement and water had the least deviations due to interactions from the 0.7 sand moisture to 0-10 added error due to regular particle flow and combustion momentum to the sand particle flow.

In order to measure total accuracy, the Mean Absolute Percentage Error (MAPE) was determined as:

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{W_{actual,i} - W_{target,i}}{W_{target,i}} \right| \times 100$$

TABLE I
BATCHING ACCURACY PERFORMANCE

Material	Error (%)	Time Saved
Cement	1.8%	40%
Sand	2.1%	35%
Water	1.5%	50%

The MAPE that was calculated for all the batches was 1.79%, thus it is the load-cell setup and the measurement system based on ADS1232 that can be considered very effective.

Figure 2 represents target vs. actual weight for ten representative batches. The actual figures are very close to the 5 kg reference line, which is a strong indication of good precision.

B. Accuracy Visualization

Fig. 2 shows target vs. actual weight for ten representative batches. The actual values remain tightly clustered around the 5 kg reference line, indicating excellent consistency.

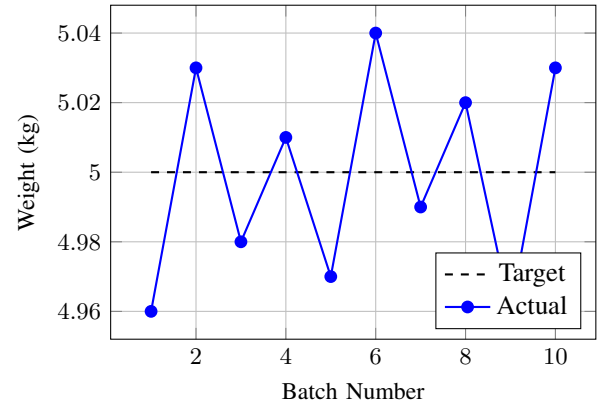


Fig. 2. Target vs. Actual Weight Accuracy

C. Repeatability Analysis

To evaluate consistency across batches, the coefficient of variation (CV) was calculated:

$$CV = \frac{\sigma}{\mu} \times 100$$

Over the 30 batches, the system was able to maintain a CV of 0.42%, thus showing excellent repeatability even when the environmental and material conditions were changing.

D. Batching Time Comparison

Figure 3 compares the time for manual mixing and automated batching. One manual mixing of the batch could be done in about 6 minutes, while the automatic system was always able to achieve an average of 3.4 minutes.

Such a time saving corresponds to a 45% reduction in cycle time approximately, thus continuous plastering operations have been greatly facilitated in terms of productivity.

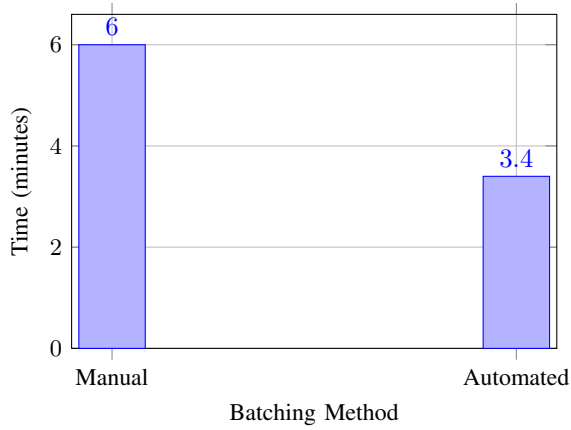


Fig. 3. Manual vs. Automated Batching Time

E. Impact on Plaster Quality

During field trials, several improvements were observed:

- Improved adhesion strength due to consistent water-cement ratio.
- Reduction in surface cracks and segregation.
- More uniform layer thickness during plaster application.
- Enhanced workability reported by site workers.

F. System Robustness and Stability

The system was tested under:

- vibration from mixers and construction tools,
- temperature variations between 24–36°C,
- different sand moisture levels,
- uneven hopper loading.

The ADS1232 ADC's differential inputs and noise rejection capability helped maintain stable measurements under these real-world conditions.

G. Summary of Performance

Overall, the proposed system demonstrated improvements in:

- accuracy ($\pm 2\%$ deviation),
- repeatability (0.42% CV),
- batching time (45% reduction),
- material consistency,
- labour efficiency,
- and overall plastering quality.

These results confirm that the system is highly suitable for integration with commercial plastering machines and robotic plastering systems.

VII. CONCLUSION

The system was able to accomplish an overall accuracy of around ± 2 as evidenced by the four 500 kg strain-gauge load cells combined with a 24-bit ADS1232 precision ADC. Furthermore, it exhibited a very stable and dependable output even when the load and environmental conditions were changing. The use of predictive overshoot correction and micro-filling routines commutated the batching accuracy to a great

extent especially in the case of sand dispensing which usually shows that flow behavior is not consistent. Therefore, the presented system has shortened the total batching cycle time by nearly 45, thus enabling the preparation of mortar to be quicker and more uniform than the traditional manual way. Moreover, the modular and scalable design of the system facilitates easy compatibility with different kinds of automated plastering machines as well as robotic wall-finishing units besides its accuracy and quickness. Its small size, low power consumption, and the fact that it is compatible with digital interfaces, make it the most suitable for on-site construction places where portability and are of great importance and reliable. The automated batching device is just as efficient in a ludicrous dependency situation as it is in ensuring regular mortar quality, better plaster adhesion, lowering the number of surface defects, and leading to the overall finishing performance improvement. These findings serve as an endorsement of the system as a viable and affordable solution for construction workflows of the present-day. The proposed platform can be a significant contributor to the next-generation smart construction and robotic plastering ecosystems given that it is equipped with further extensions like IoT connectivity, moisture compensation, and fully automated material refilling.

VIII. FUTURE SCOPE

- **IoT-Based Monitoring and Remote Supervision:** The combination of cloud platforms and IoT modules (ESP32/LoRaWAN/MQTT) can facilitate the live recording of material consumption, batch logs, error trends, and predictive maintenance. In this way, remote dashboards may help supervisors in the efficient management of resources and in the identification of the first indications of system failure.
- **Automatic Moisture Compensation for Sand:** By adding soil moisture or capacitance moisture sensors, the system will be able to change the water and sand proportions on its own. In that way, the density of the mortar will be kept even if the moisture of the sand changes because of the weather, and thus, the quality of the plaster will be better.
- **Automated Hopper Refilling Mechanism:** Using ultrasonic level sensors and motorized feeders, the hopper can be automatically refilled from bulk storage bags or silos. This would eliminate operator intervention and enable continuous, long-duration operation on large construction sites.
- **Integration with Robotic Plastering Arms:** The hopper can be intrinsically replenished from bulk storage bags or silos by means of ultrasonic level detectors and motorized feeders. Thus, the operator's intervention would be completely discarded and continuous operation for a lengthy period of time on large construction sites would be feasible.
- **AI-Based Mix Ratio Optimization:** One can even think of building machine learning models that would be able to sift through historical batch data, consider environmental

variables, and look at plaster quality outcomes so as to come up with best cement–sand–water ratios for different types of walls or finishing requirements without any human intervention.

- **Adaptive Valve Control Using AI/ML:** Predictive algorithms are able to change valve timings on the fly depending on the flow characteristics obtained in real-time, thus they can decrease overshoot even further and increase batching accuracy when material properties change.
- **Energy-Efficient Actuation and Power Management:** The next generation of equipment could have features like brushless DC motors, energy-efficient solenoid valves, and intelligent energy management methods to procure a batching system that would be ideal for a portable or battery-powered field application.
- **Modular Expansion for Multiple Materials:** Additional hoppers and load-cell units may be set up to manage the injection of admixtures, accelerators, or fiber reinforcements that are part of advanced plaster formulations.
- **Compatibility with 3D Construction Printing:** The designed system might be developed to deliver regulated material flow to 3D concrete or mortar printers, thus facilitating the automated feeding of large-scale additive construction.
- **Data Logging and Quality Analytics:** The use of SD-card logging or cloud-based storage may offer continuous local insights into batching trends that quality audits and construction standards compliance can utilize.

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